

Project Design Document

What problem are we solving?

The company is experiencing an increase in major defects across product lines that exceed Acceptance Quality Limit (AQL) threshold of 2.0% Leadership wants to understand the operational impact of these defects.

What is the proposed solution?

Review the inventory and shipping records for all product SKUs. Analysis will solve for the following business questions:

1. What is the operational impact of quality control failures?
2. Which product lines carry the highest defect liability?
3. Which suppliers present the greatest fulfillment bottleneck?

What is the project's scope?

1. Manufacturing

- **Defect Rate:** Measure production quality and scrap rate to highlight issues in raw materials or manufacturing steps. Use Defect rates and Inspection results. Group by Product type.
 - Use Pareto bar chart
- **Manufacturing Cycle Time:** Measure how long it takes to convert raw material into finished goods per unit. Use Manufacturing lead time and Production volumes.
 - Use Box Plot
- **Manufacturing Cost per Unit:** Measure production efficiency and unit margins. Use Manufacturing costs and Production volumes.
 - Use Stacked Bar Chart

2. Inventory

- **Inventory Turnover & Sell-Through Rate:** Track how fast stock moves versus sitting idle. Use Number of products sold, Stock levels, and Availability.
 - Use Dual-Axis Line Chart
- **Supplier Lead Time Variance:** Measure supplier delivery lead times versus expected delivery schedules. Use Lead times and Supplier name.
 - Use Scatter Plot Chart

- **Average Order Quantity (AOQ):** Track using Order quantities grouped by SKU or Product Type. Helps optimize reorder points and batch sizes.
 - Use Horizontal Bar Chart

3. Logistics

- **On-Time Delivery (OTD):** Measure efficiency across each carrier and route. Use Shipping times alongside expected delivery schedules.
 - Use Donut Chart
- **Cost per Shipped Unit:** Measure freight efficiency across each transportation mode and shipping carrier. Use Shipping costs and Order quantities.
 - Use Heatmap
- **Carrier & Route Cost-Efficiency:** Identify expensive transportation modes. Use Costs grouped by Transportation modes and Routes.
 - Use Bubble Chart

4. Financial

- **Gross Profit Margin per Product:** Measure product profitability. Use Revenue generated, Manufacturing costs, and Shipping costs for each SKU per Product Type.
 - Use Bar Chart (or possibly Waterfall Chart)
- **Average Revenue per Unit (ARPU):** Measure pricing across each market segment. Use Revenue generated, Number of products sold, and Customer demographics.
 - Use Treemap

What is the dashboard design?

The dashboard design will include tabs for each metric category (Manufacturing, Inventory, Logistics, and Financial).

Will include global filters to allow filtering by Product Type, Transportation Mode, Supplier Name, and Route.

Create BANs for executive-level KPIs (On-Time Delivery (OTD), Avg Lead Time, Defect Rate, Inventory Turnover Ratio, Sell-Through Rate, Inventory Value).